



BEST PRACTICE SERIES

Improved 3500 Engine Exhaust Manifold Salvage

Maintenance and Repair Application Component Renewal (CRC) Component Life Management Component Rebuild MARC Management

3500 Engine Exhaust Manifold Salvage **Error!**
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1.0 Introduction

Significant cost savings can be achieved by salvaging the ends (seals area) of certain 3500 engine exhaust manifolds.

2.0 Best Practice Description

The ends of the manifolds are machined down, a fabricated repair sleeve is installed, and the manifold can return to service.

3.0 Implementation Steps

CAUTION: SAFETY PROCEDURES FOR THE SHOP EQUIPMENT INVOLVED MUST BE FOLLOWED.

- Clean the exhaust manifold using a shot-peen or sandblast method.
- Inspect the manifold for cracks. If a crack is found, salvage of the manifold is not recommended, as the cracks may continue to propagate.
- Machine the end of the manifold to $4.374 \pm .001$ inches, 1.750 inches long.



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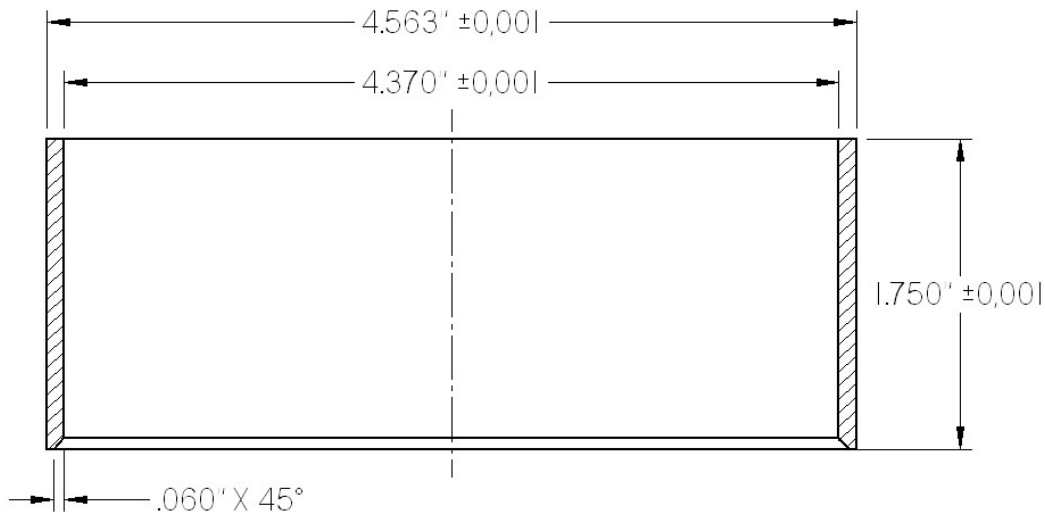
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- Fabricate a sleeve from SAE 1040 grade mechanical tubing, DOM, 4.370" ID by 4.563" OD by 1.750" length with an inside chamfer of 45 degrees by .06 long. The inside diameter should be about 4.370 as delivered.



- Heat the fabricated sleeve to approximately 250 degrees F.
- Before dropping the sleeve onto the manifold some "High Temperature Manifold Sealer" 2P-2333 is applied to it by means of a brush.
- Drop the heated sleeve onto the machined exhaust manifold using heat resistant gloves. The sleeve material should provide .003 to .004 inches interference fit.

4.0 Benefits

- The finished manifolds can be mixed along with new manifolds on the same engine bank, because the centerlines have not changed.
- To reduce turnaround time for an engine rebuild, sufficient quantities of spare manifolds can be inventoried.
- The manifold can be sleeved additional times, if inspection criteria about cracking are met.

5.0 Resources Required

- 22-inch swing lathe with cutting tools.
- Oven.

6.0 Supporting Attachments / References

See publication SEPD0611, page 36, "A new exhaust seal and a new exhaust manifold sleeve are used in the exhaust manifold groups."

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7.0 Related Best Practices

See Best Practice number 0907-4.5-1091

8.0 Acknowledgements

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